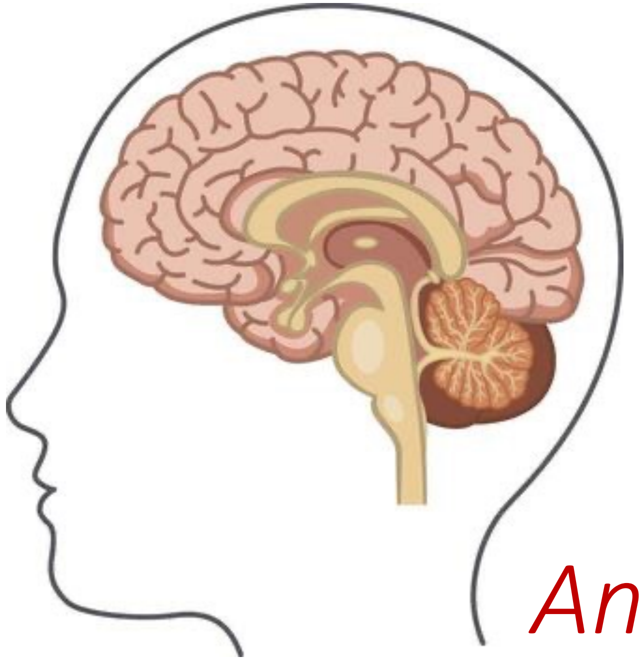
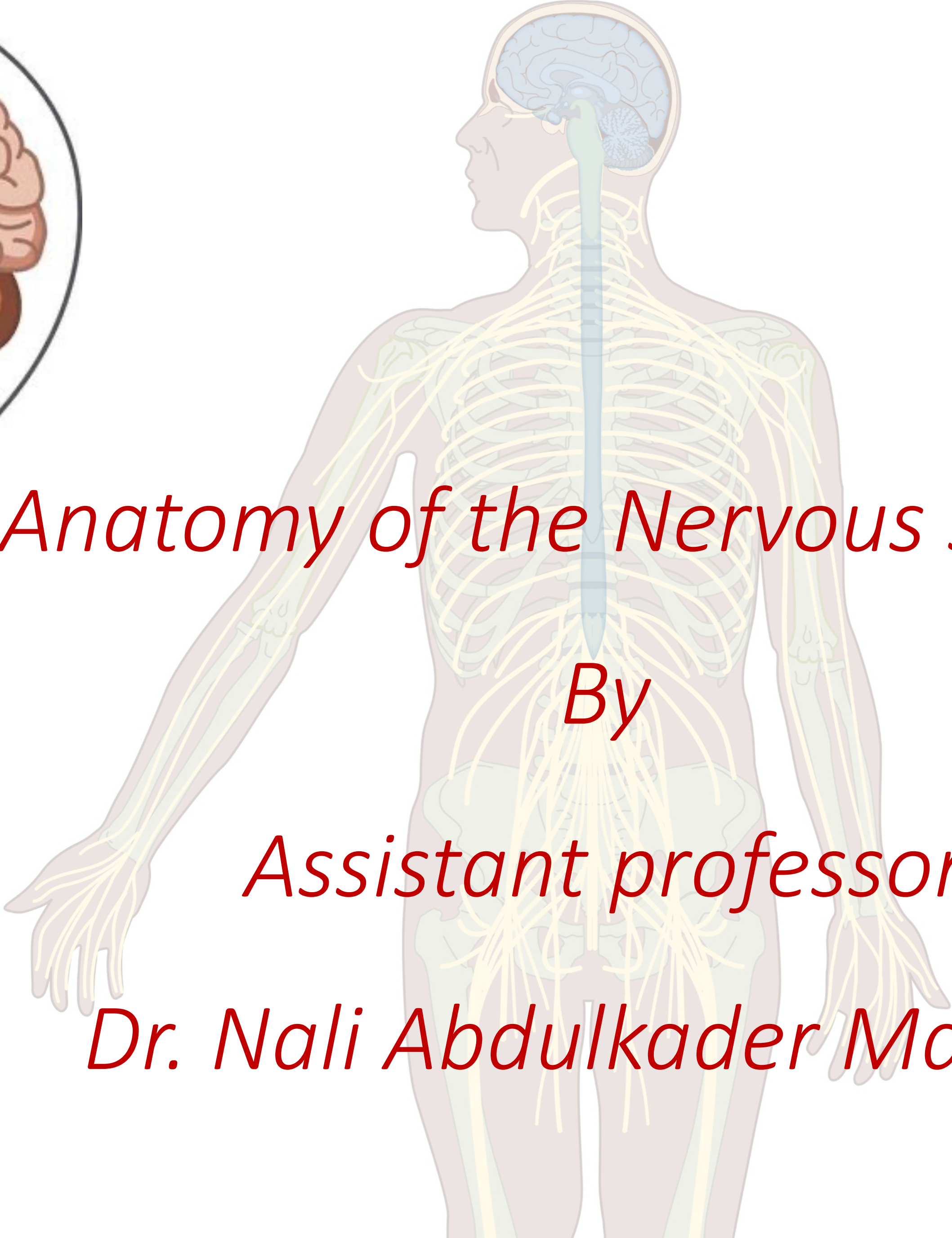
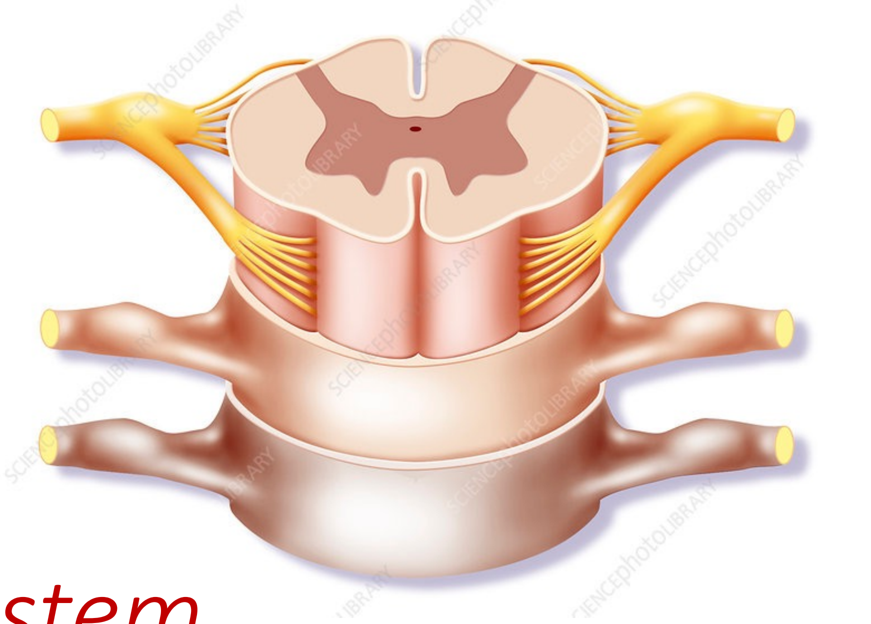




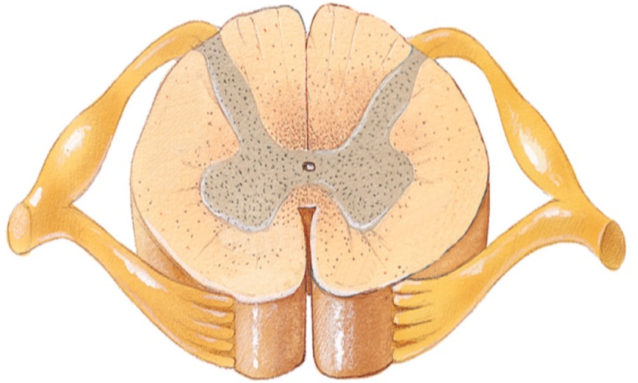


Anatomy of the Nervous system

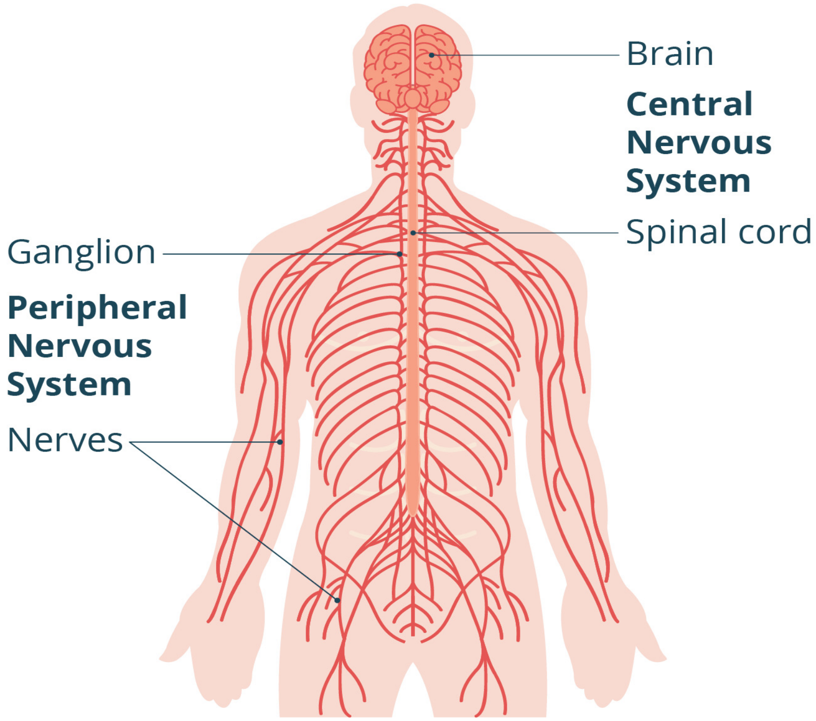
By

Assistant professor

Dr. Nali Abdulkader Maaruf



Nervous system

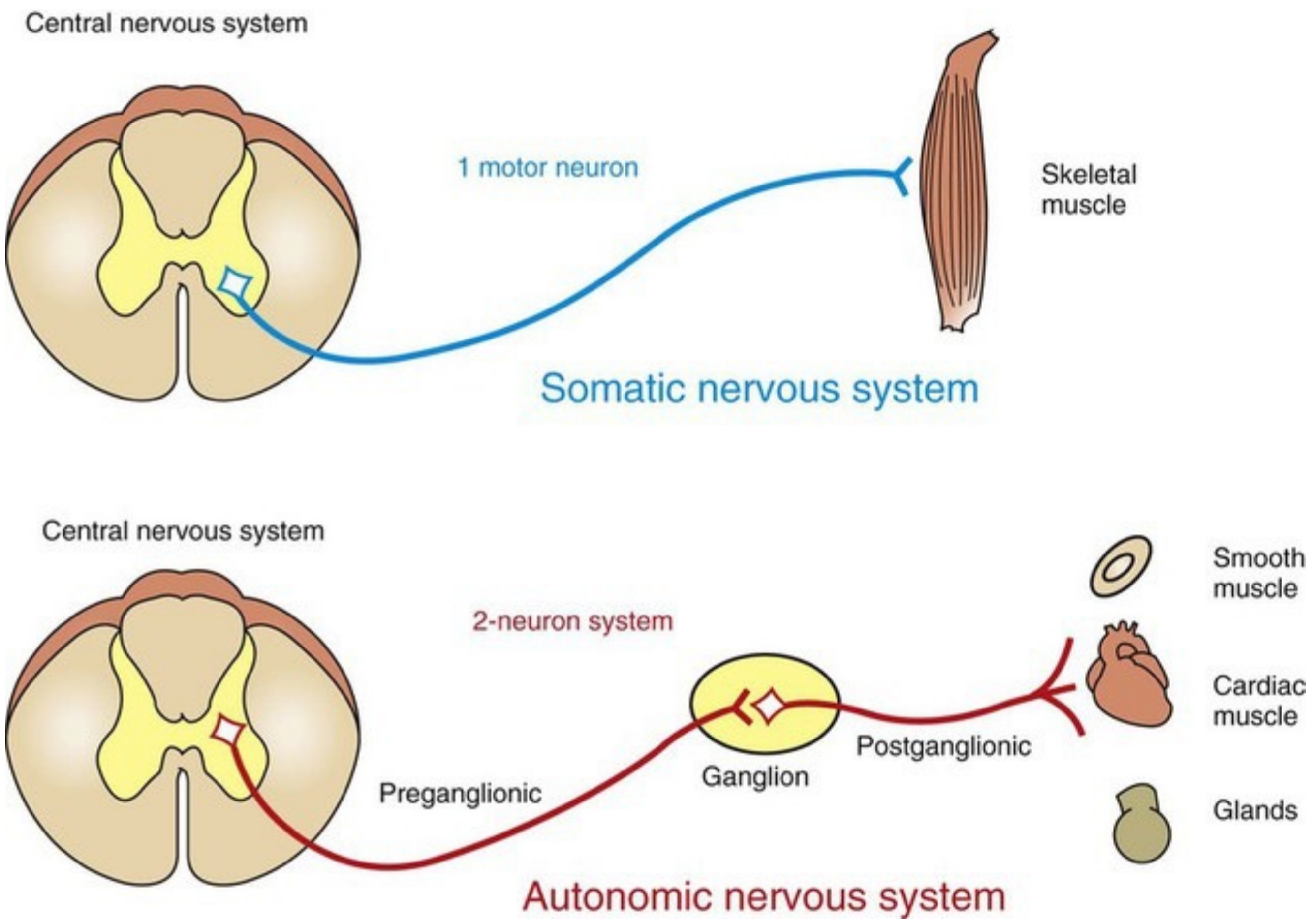


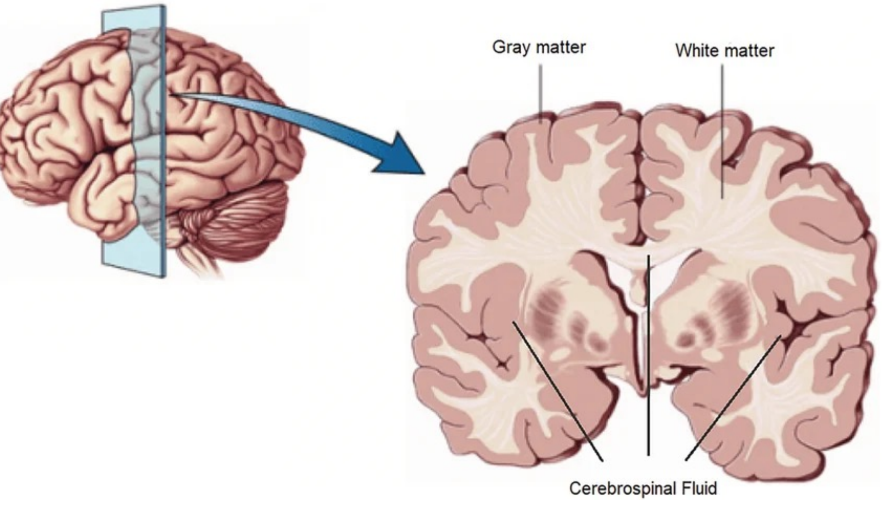
- The nervous system is a complex system formed by millions of specialized cells called neurons, supported by neuroglial (glial) cells.
- It functions to receive, process, and transmit information
- *The nervous system together with the endocrine system controls and integrates the activities of the different parts of the body.*
- The nervous system has two main parts:
  - Central Nervous System (CNS) – made up of the brain and spinal cord they protect by meninges
  - Peripheral Nervous System (PNS) – made up of spinal nerves (31 pairs) and cranial nerves (12 pairs)

Functionally

- Functionally the nervous system can be further divided into the

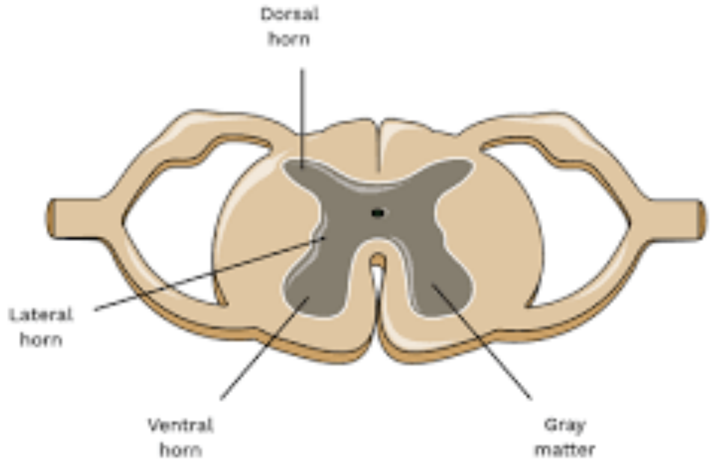
1. **Somatic nervous system** which controls voluntary activities and
2. **Autonomic nervous system** which controls involuntary activities.

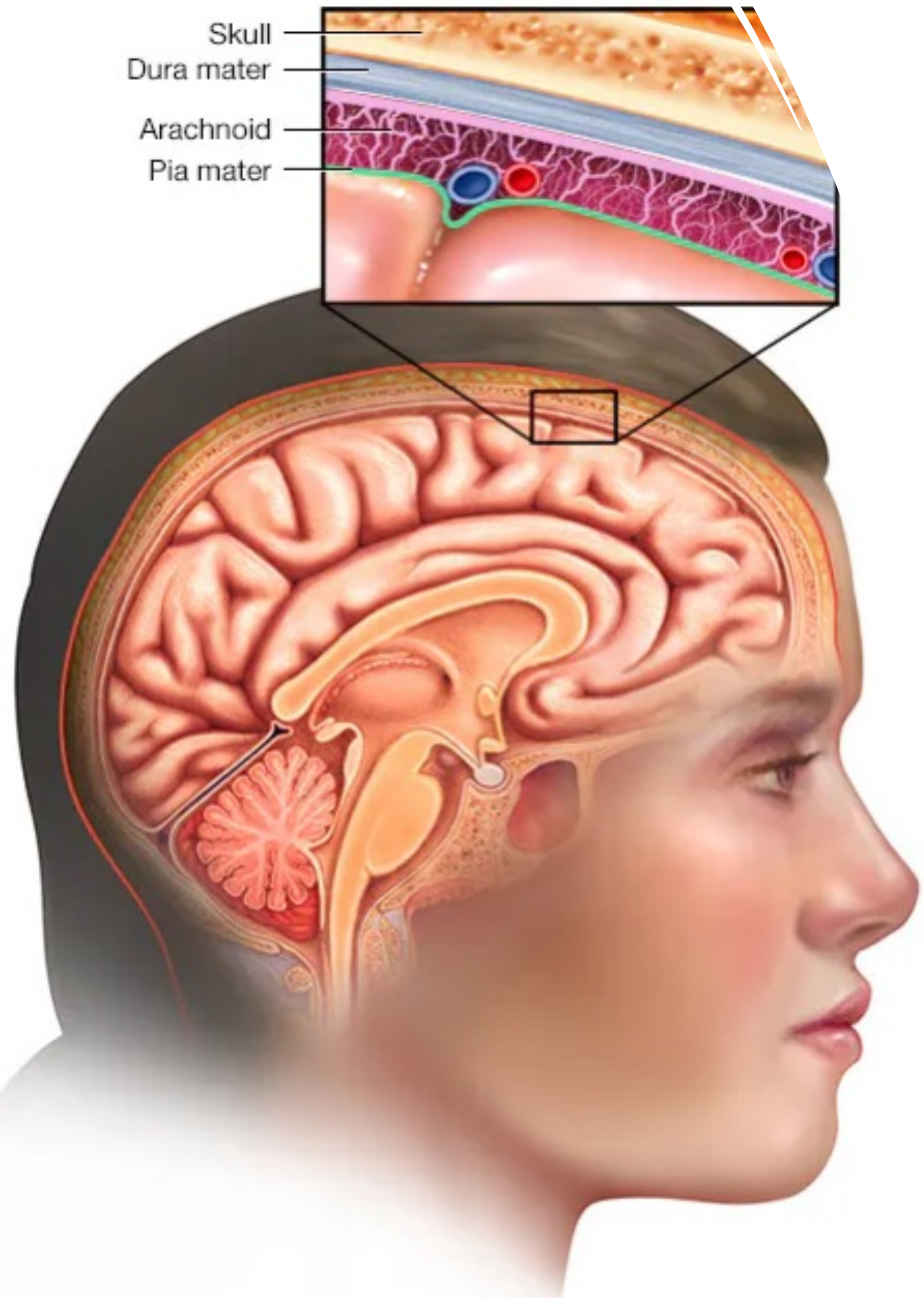




Central Nervous System

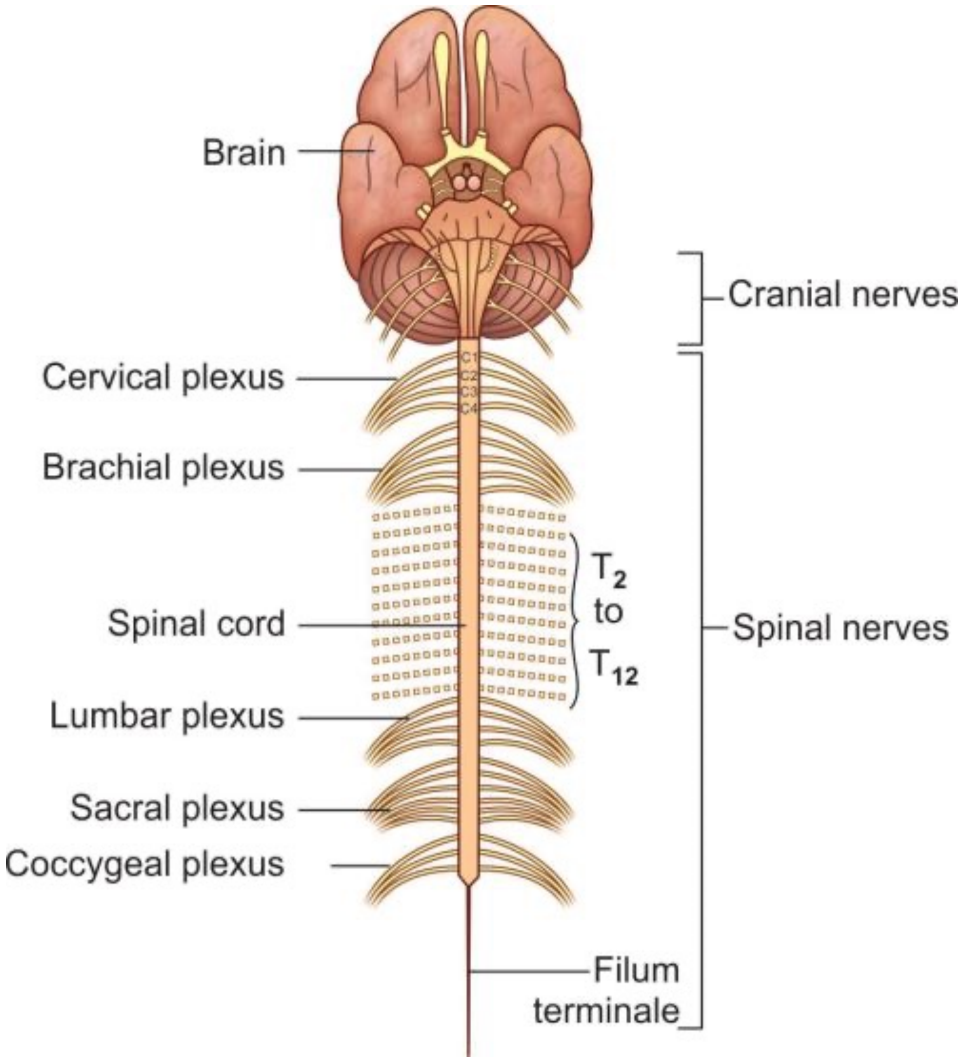
- It consists of the brain and spinal cord.
- The brain and spinal cord are composed of gray matter and white matter.
- The nerve cell bodies constitute the gray matter.
- The axons form the white matter.
- Nucleus is composed of a group of cell bodies with same function inside the central nervous system.
- A bundle of axons inside the CNS is named as tract.





Covering of the central nervous system

- **The brain** is inside the cranial cavity by which it is well protected.
- **The spinal cord** lies within the vertebral canal by which it is well protected.
- Brain and spinal cord have three coverings ,from outer to inner they are dura matter, arachnoid matter and pia matter. (the meninges)



Peripheral nervous system

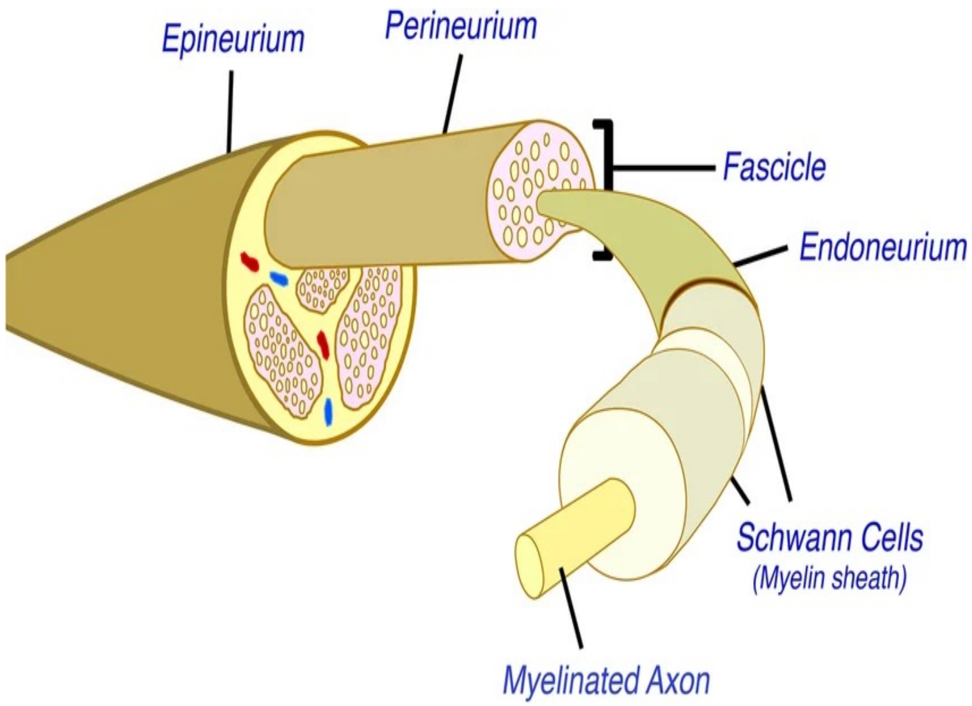
- Is composed of

A.) 12 pairs of cranial nerves. (they arise from the brain)

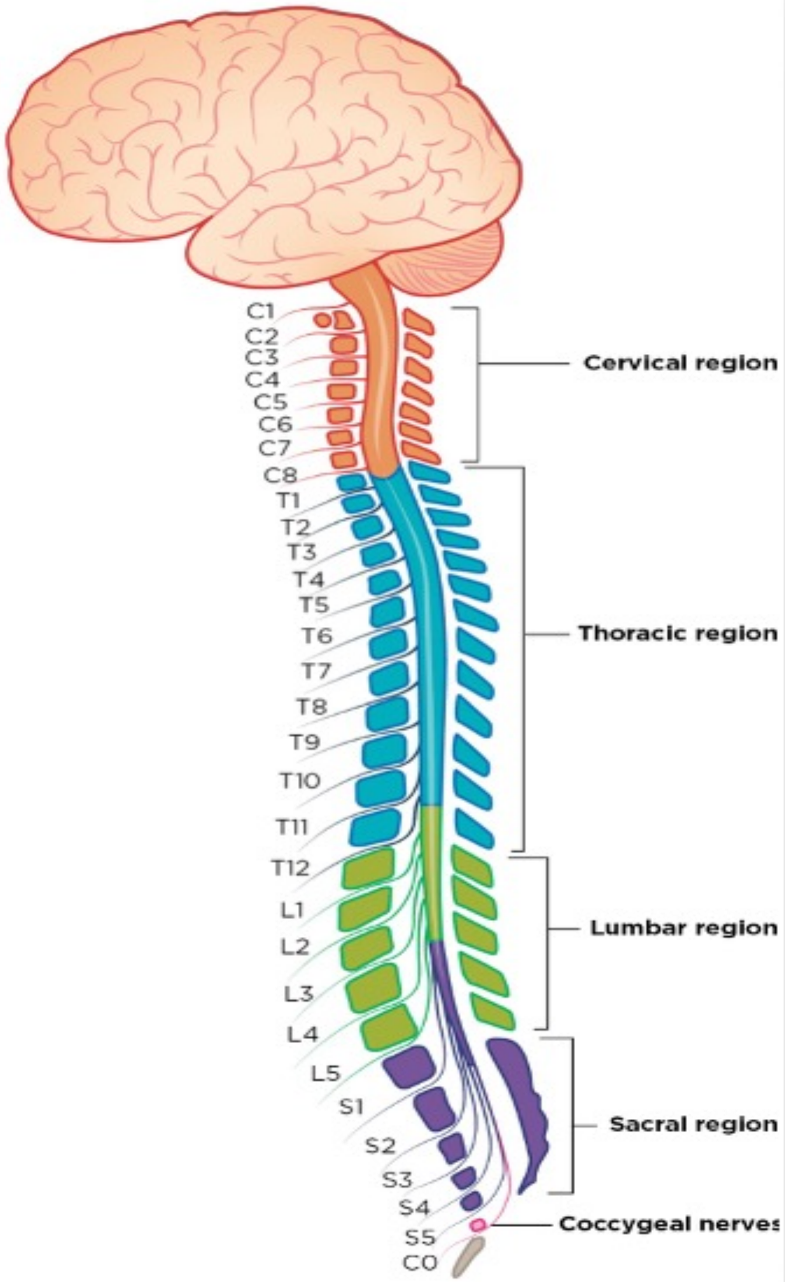
B.) 31 pairs of spinal nerves. (they arise from the spinal cord)

C.)Ganglia(which are composed of collections of nerve cell bodies outside the CNS)

Structure of the peripheral nerves

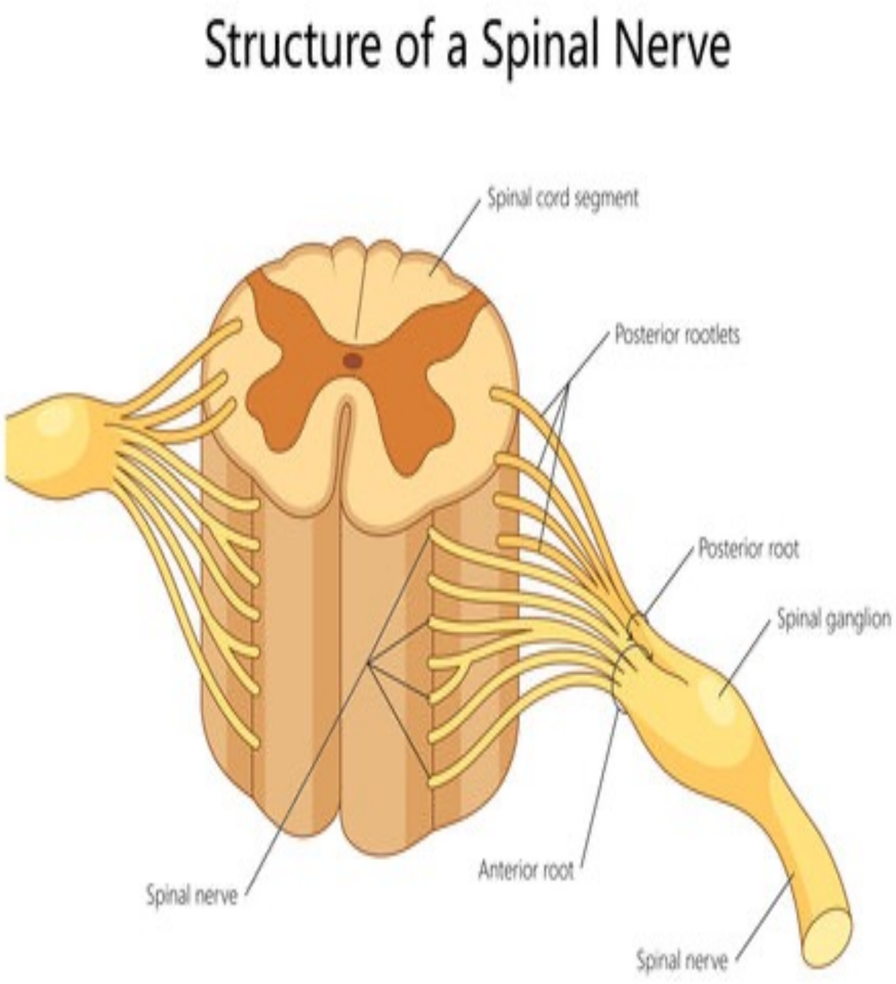


- A peripheral nerve (either cranial or spinal) is composed of bundles of nerve fibers (axons) whose cell bodies lie within the gray matter of CNS or in a ganglion.
- In any nerve, each nerve fiber (axon) is surrounded by series of schwann cells which either myelinate it (provide it with a myelin sheath) or do not.(in unmyelinated nerve fibers.
- myelin sheath is derived from the plasma membrane of the shcwann cells.



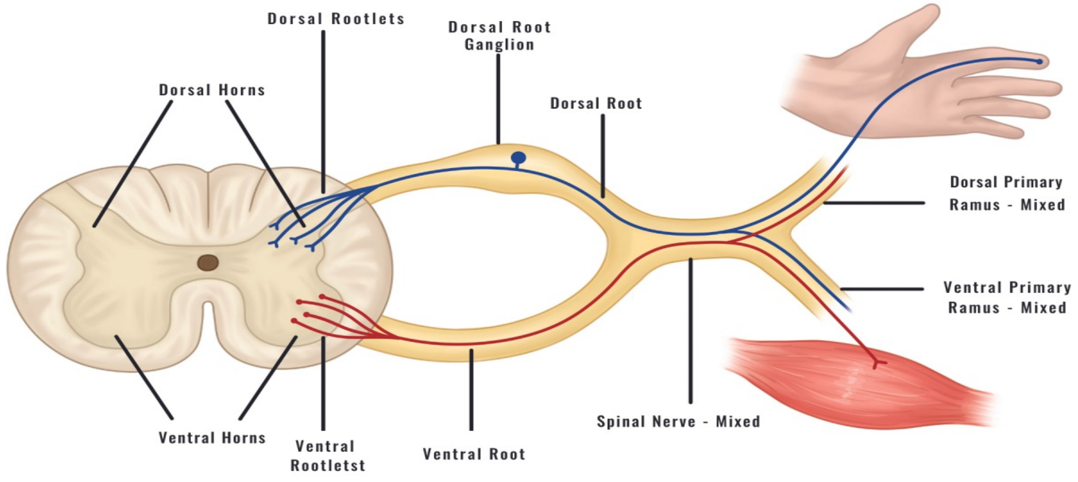
Spinal Nerves

- The spinal cord is composed of 31 spinal segments which are 8 cervical 12 thoracic, 5 lumbar, 5 sacral and one coccygeal spinal segments.
- Each segment gives rise to a pair of spinal nerve on each side.
- In cross section the spinal cord have a peripherally located white matter and at the center there is an H shaped gray matter.



Spinal Nerves

- Thus there are two anterior horns (composed of motor nuclei)and two posterior horns (composed of sensory nuclei)
- In addition to these, there are lateral horns in the spinal segments T1 to L2. (nucleus of sympathetic preganglionic neurons)
- Also there are lateral horns in S2 S3 and S4 segments (nucleus of parasympathetic preganglionic neurons)

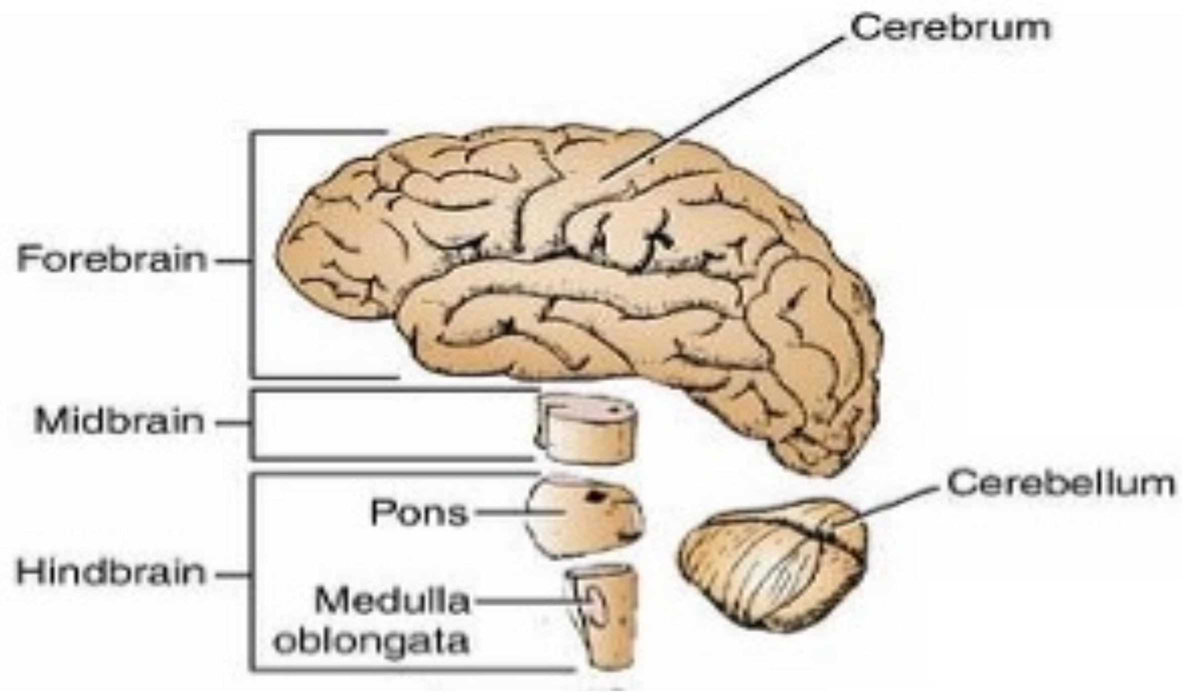


Formation of the Spinal Nerves

- Each spinal nerve arise by two roots, anterior root which is the motor (efferent) root, and posterior root which is sensory root. (afferent)
- In the dorsal root there is dorsal root ganglion which contains sensory neurons.
- The two roots unite to form the trunk of the spinal nerve which immediately divide into ventral ramus and dorsal ramus of the spinal nerve.
- The ventral ramus supplies anterior and lateral parts of the trunk and the limbs.
- The dorsal ramus supplies the posterior part of the trunk.

Brain

- Is the largest part of CNS, locates in the cranial cavity and is about 1.3–1.4 kg (≈3 lbs).
- It contains ~86 billion neurons and trillions of connections (synapses).
- The brain accounts for about 2% of body weight but uses ~20% of the body's oxygen and energy.
- The brain is 60% fat, making it the fattiest organ in the body

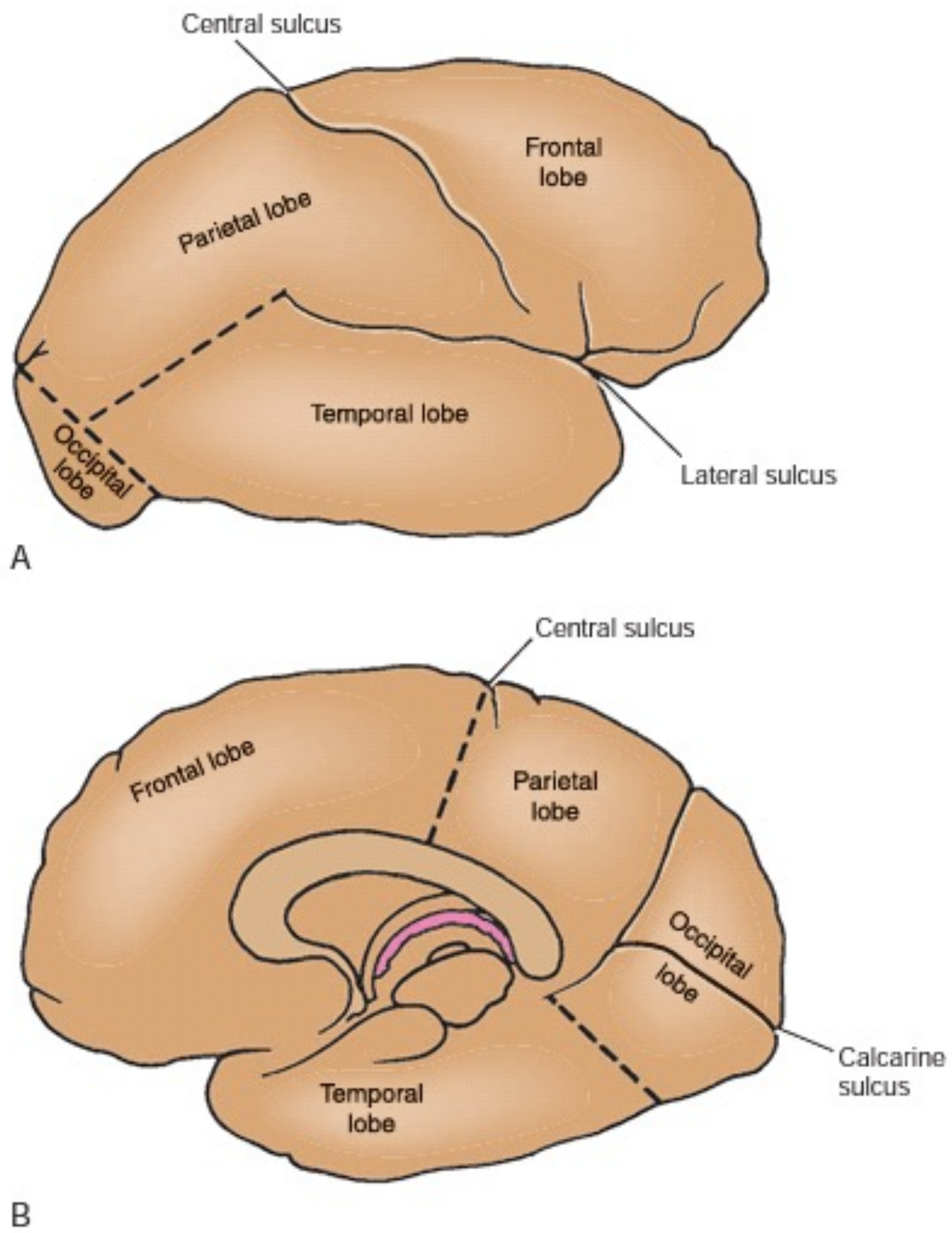


- The brain consists of
  - Forebrain: *Cerebrum and Diencephalon*
  - Midbrain
  - Hindbrain: *Medulla Oblangata, Pons and Cerebellum*

Cerebrum

The cerebral cortex, the outermost portion of the cerebral hemisphere, is a gray matter structure characterized by many convolutions that greatly increase its surface area.

The bulges or eminences referred to as gyri, and the spaces separating the gyri are called sulci.

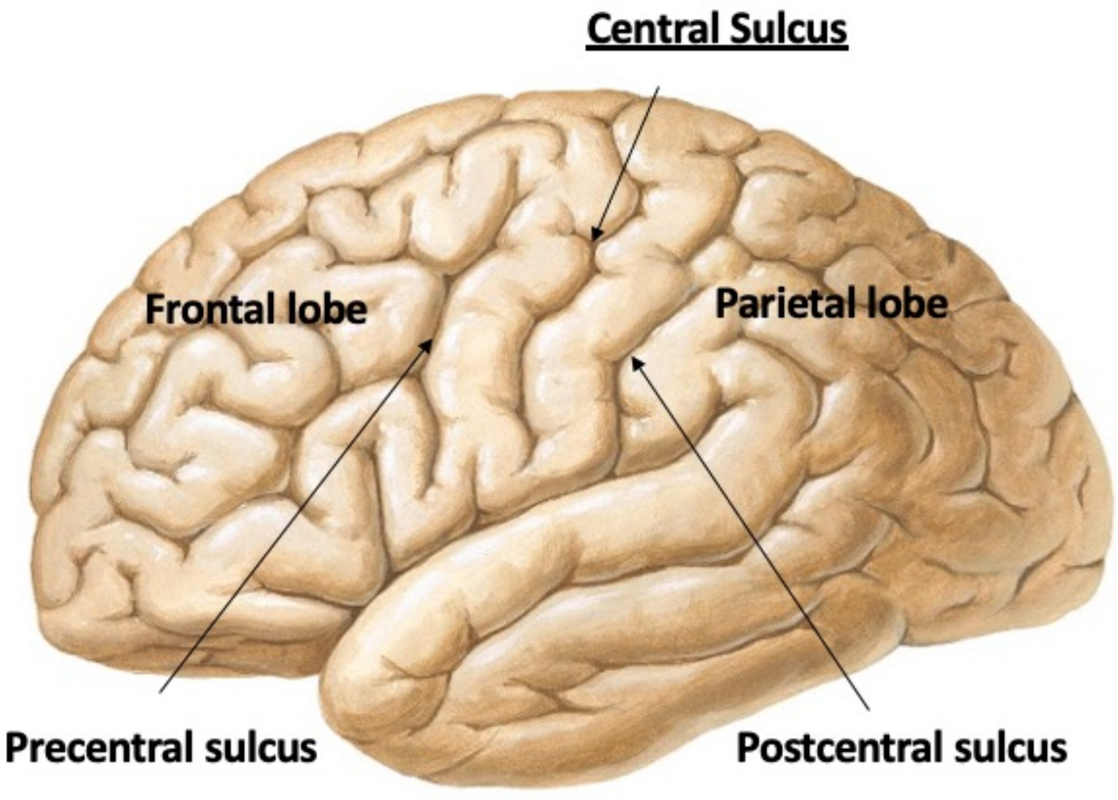


Important sulci

1. Central sulcus
2. Lateral sulcus

Lobes:

1. Frontal
2. Parietal:
3. Temporal:
4. Occipital:



Major gyri and important cortical areas:

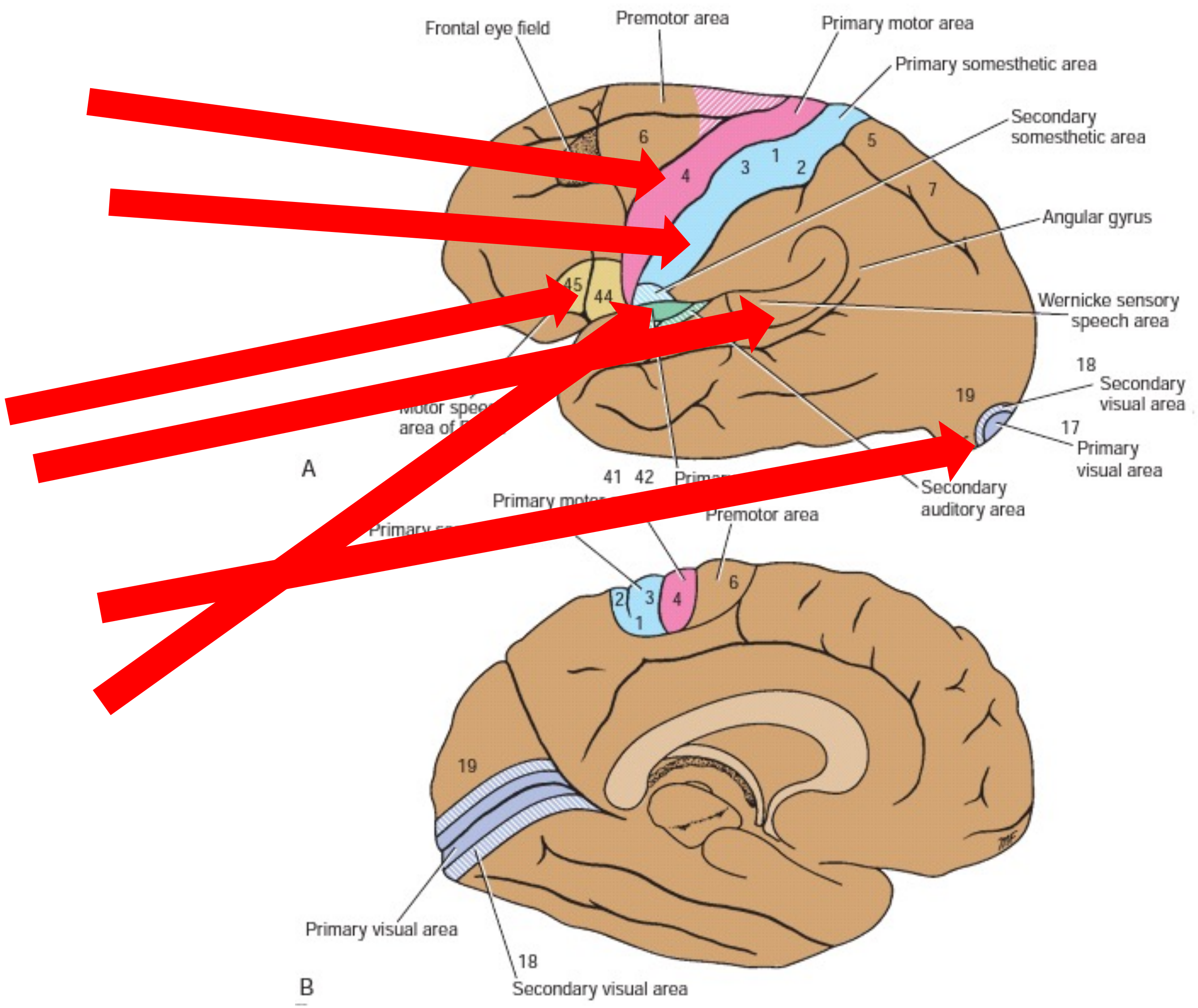
1. precentral gyrus (is motor area)
2. Post central gyrus (sensory area)
3. Near the calcarine sulcus(visual area)
4. Brocas area (motor speech area)
5. Wernicke area (sensory speech area)

Important cortical areas

- Motor area
- Sensory area
- Speech areas

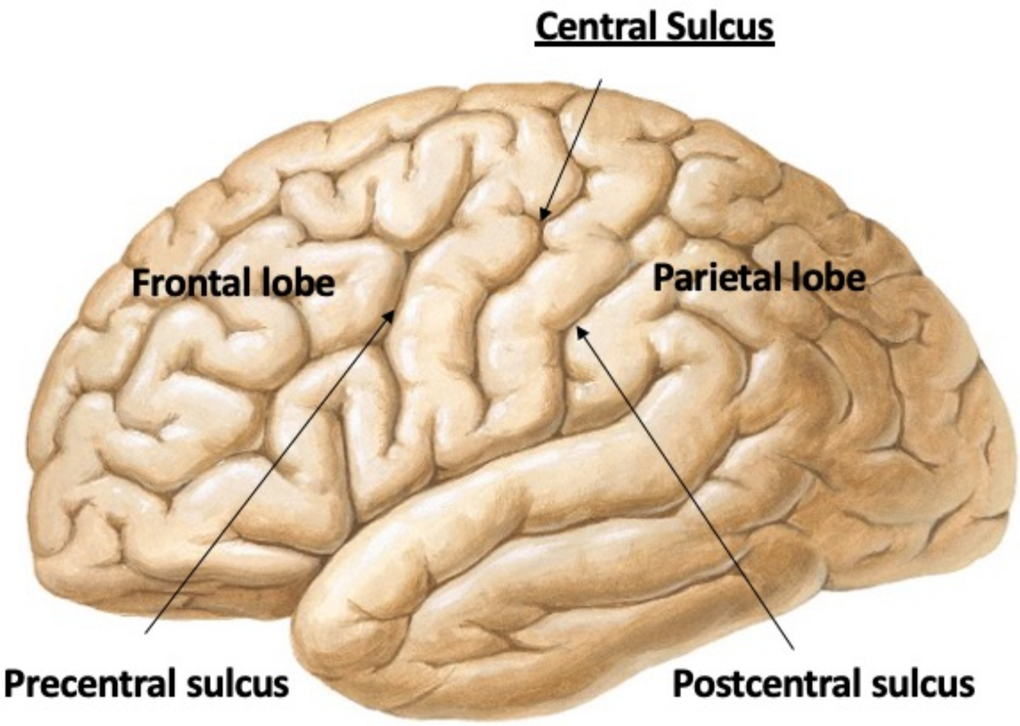
  Broca

  Wernickie
- Vision area
- Hearing area



Lesions affecting specific cortical areas

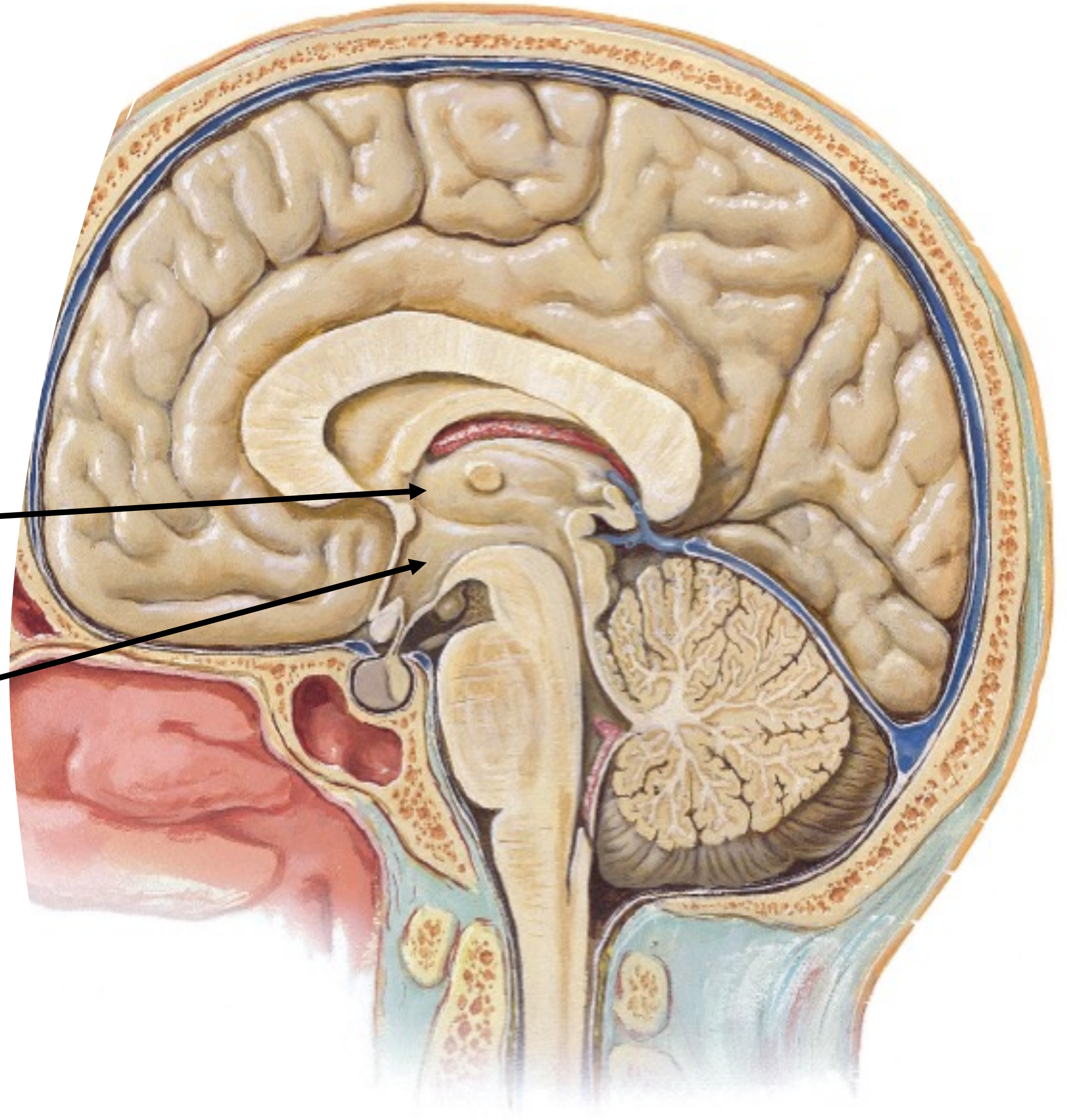
- **Motor area:** contralateral(opposite side) weakness of the upper and lower limbs
- **Sensory area:** loss of sensation on the contralateral upper and lower limbs
- **Broca's area**: aphasia(understands speech but can't speak)
- **Wernickie's area**: aphasia(can speak but not understand speech)
- **Vision area**: blindness



- *More than 90% of the adult population is right-handed and, therefore, is left hemisphere dominant.*

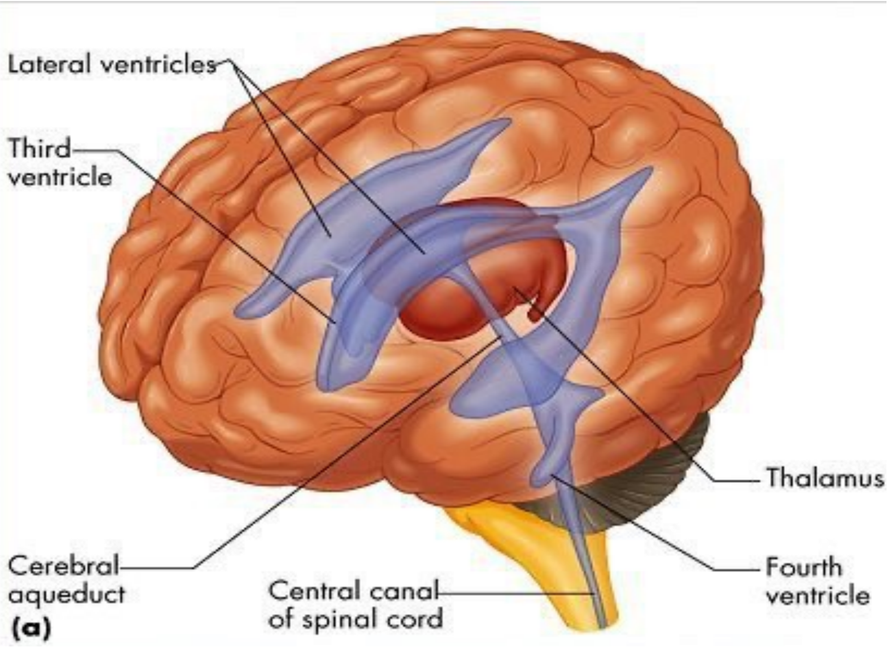
Anatomy of Brain: deep structures

- **Thalamus:** situated on lateral sides of third ventricle
- Function: sensory relay station
- **Hypothalamus:** situated below thalamus, extends from optic chiasm to mammillary bodies
- Function: control of body temperature, eating, drinking, autonomic functions….etc



Ventricular System of the Brain

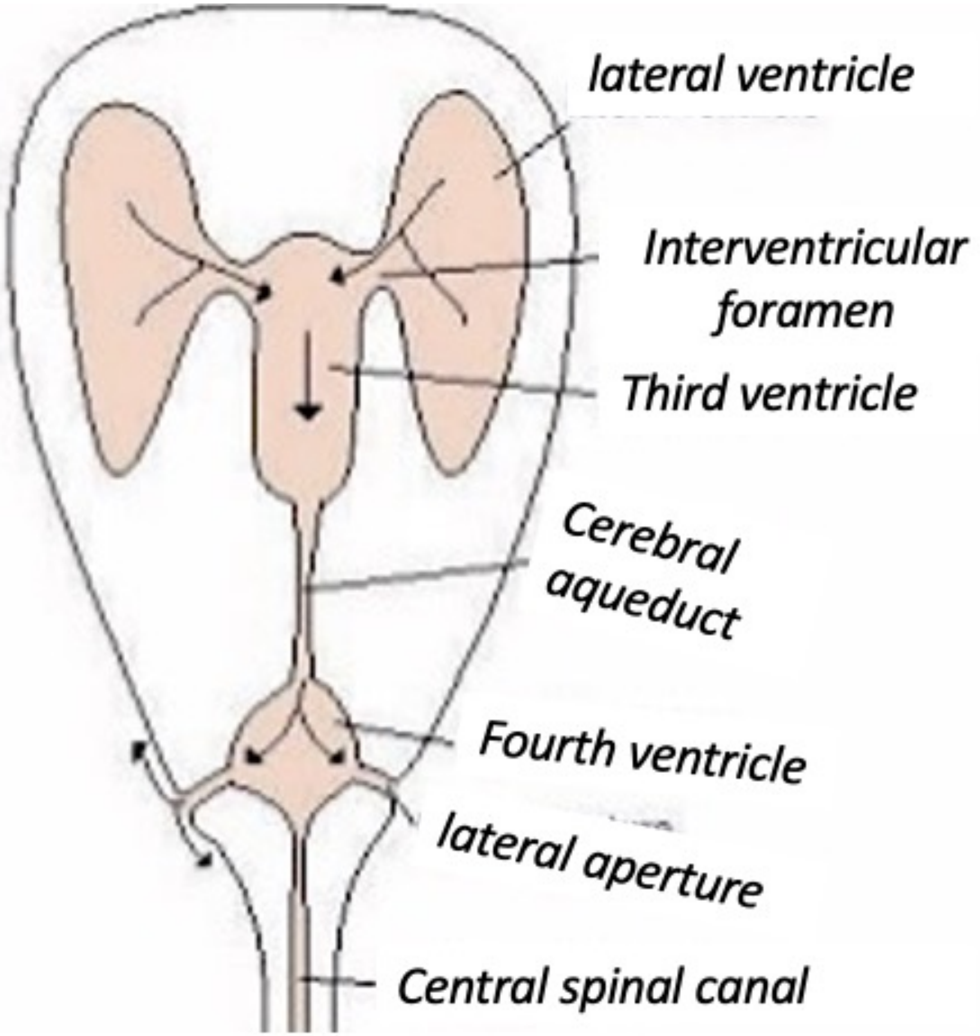
- 2 Lateral Ventricles
- Third Ventricle
- Fourth ventricle
- The Cerebrospinal Fluid(CSF) is a clear, colorless liquid that fills the ventricles of the brain and bathes the external surface of the brain and spinal cord.
- It is formed from the choroid plexus within the ventricles



Communications between ventricles

They are connected by

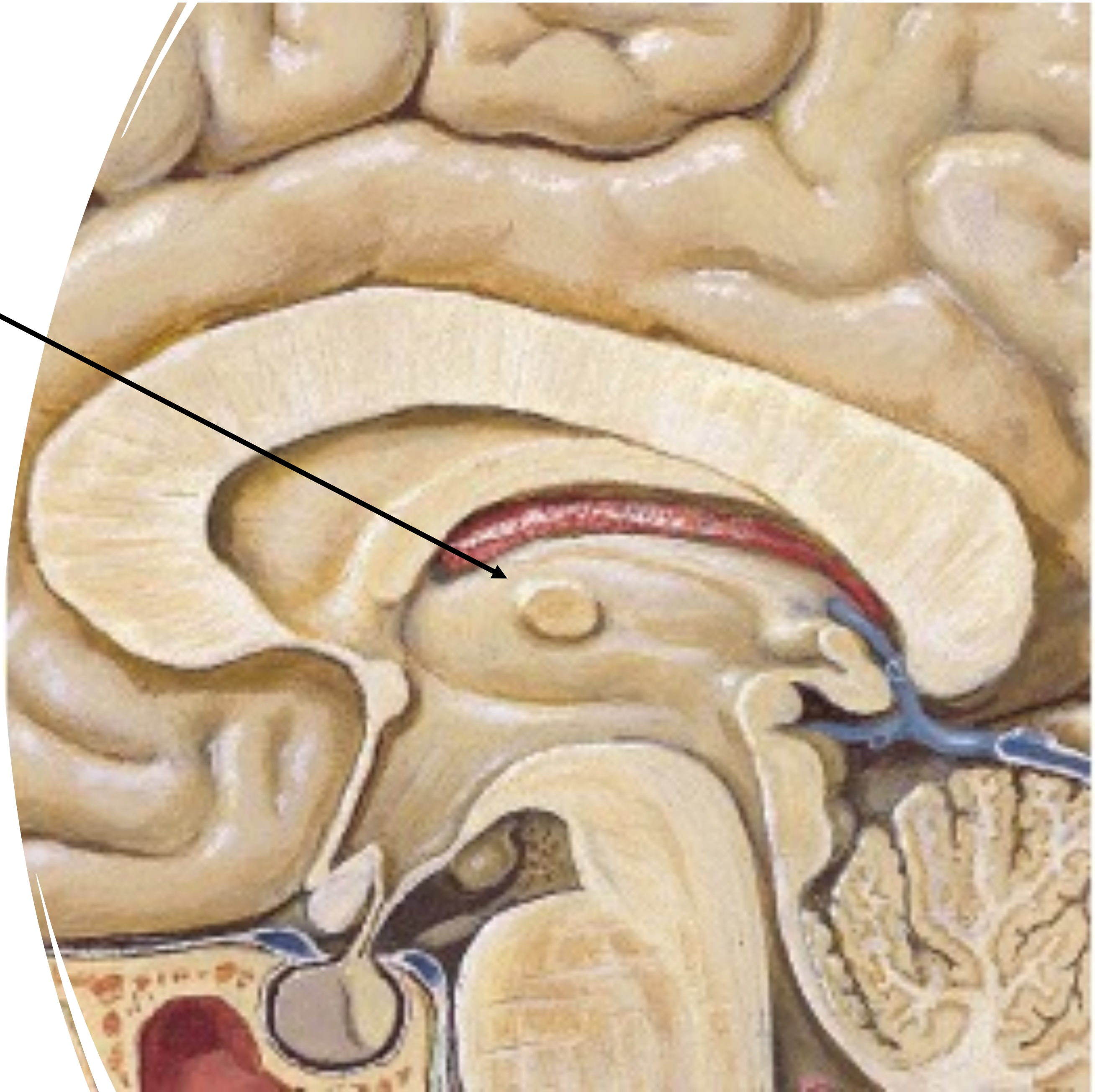
- The foramen of Monro (lateral → third),
- Cerebral aqueduct (third → fourth), and
- The foramen of Magendie and Luschka (fourth → subarachnoid space / cisterna magna).



- Obstruction of Monro → unilateral hydrocephalus
- Aqueductal stenosis → severe obstructive hydrocephalus
- Blockage of Magendie and Luschka → enlargement of the 4th ventricle

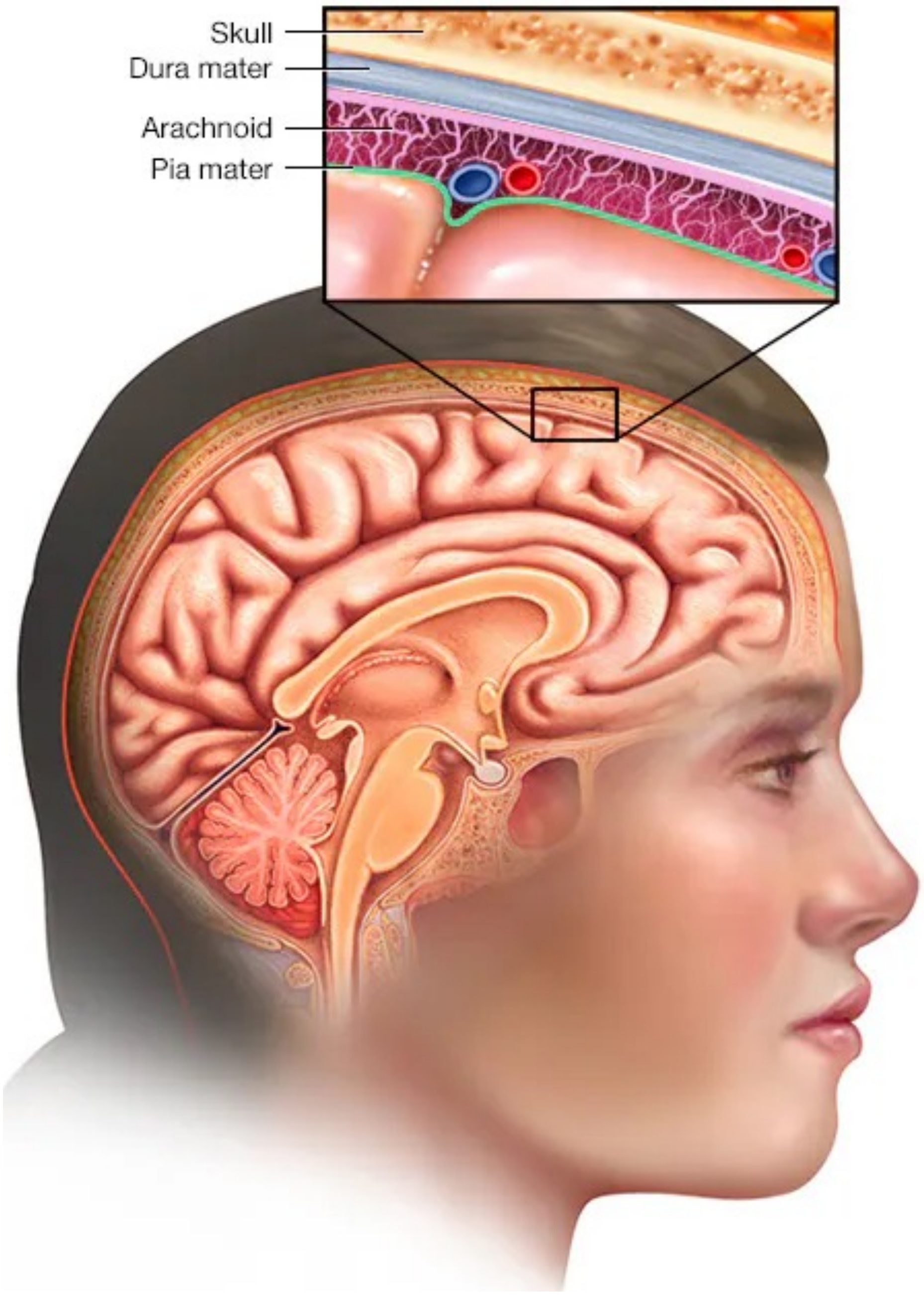
The thalamus

- The thalamus is a large, egg-shaped mass of gray matter that forms the major part of the diencephalon.
- There are two thalami, and one is situated on each side of the third ventricle and connected by (interthalamic adhesion)



Functions of CSF:

1. Cushions and protects the central nervous system(CNS) from trauma
2. Provides mechanical support for the brain
3. Assists in the regulation of the contents of the skull
4. Nourishes the CNS
5. Removes metabolites from CNS

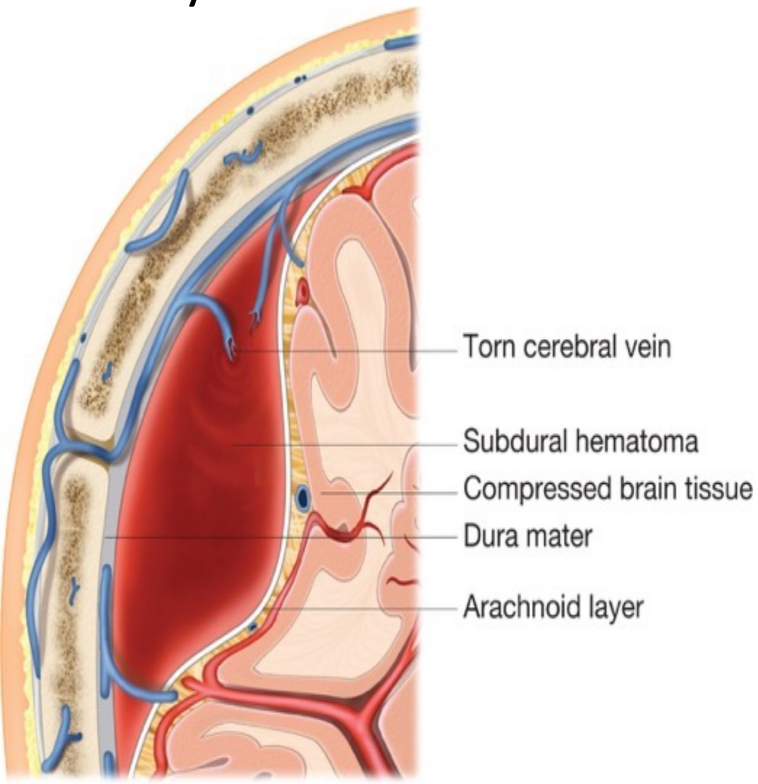


Meningeal Linings of the Brain:

1. Dura mater
2. Arachnoid
3. Pia mater

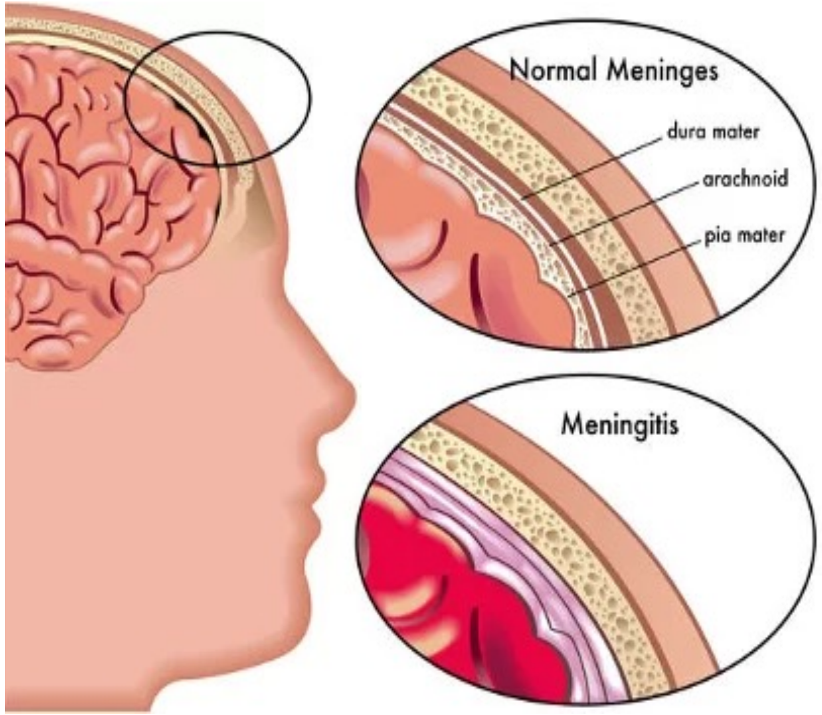
Subdural Hemorrhage (SDH)

- Bleeding between the dura and arachnoid mater.
- Rupture of bridging veins, often after head injury or in elderly.



Meningitis

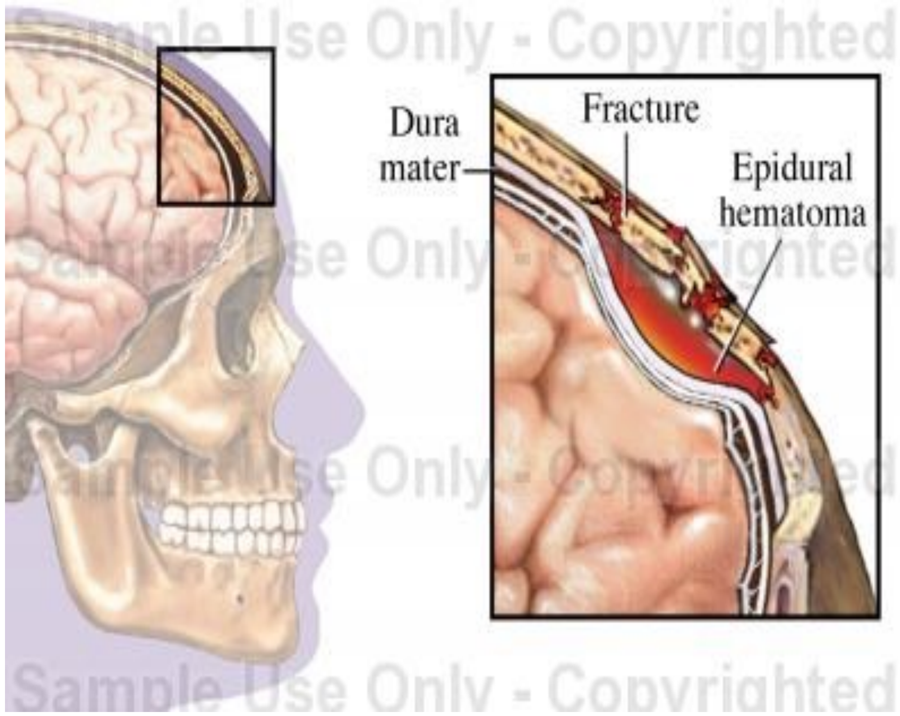
- Inflammation of the meninges (dura, arachnoid, pia).
- **Infection** (bacterial, viral, or fungal)



Epidural Hematoma (EDH)

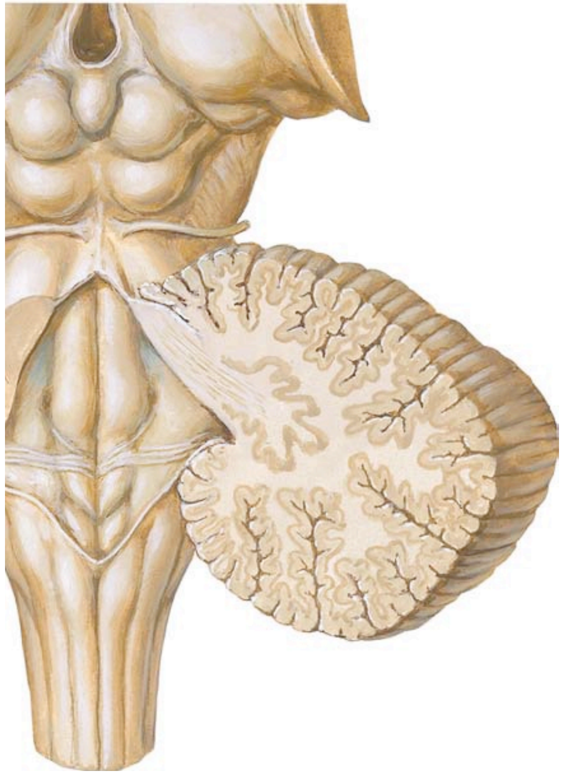
- Bleeding between the skull and dura mater.

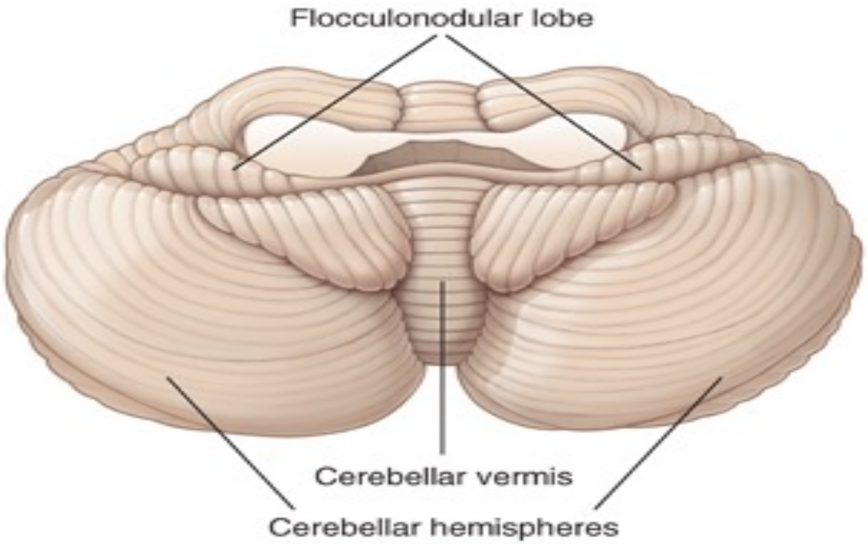
Trauma → usually middle meningeal artery rupture.

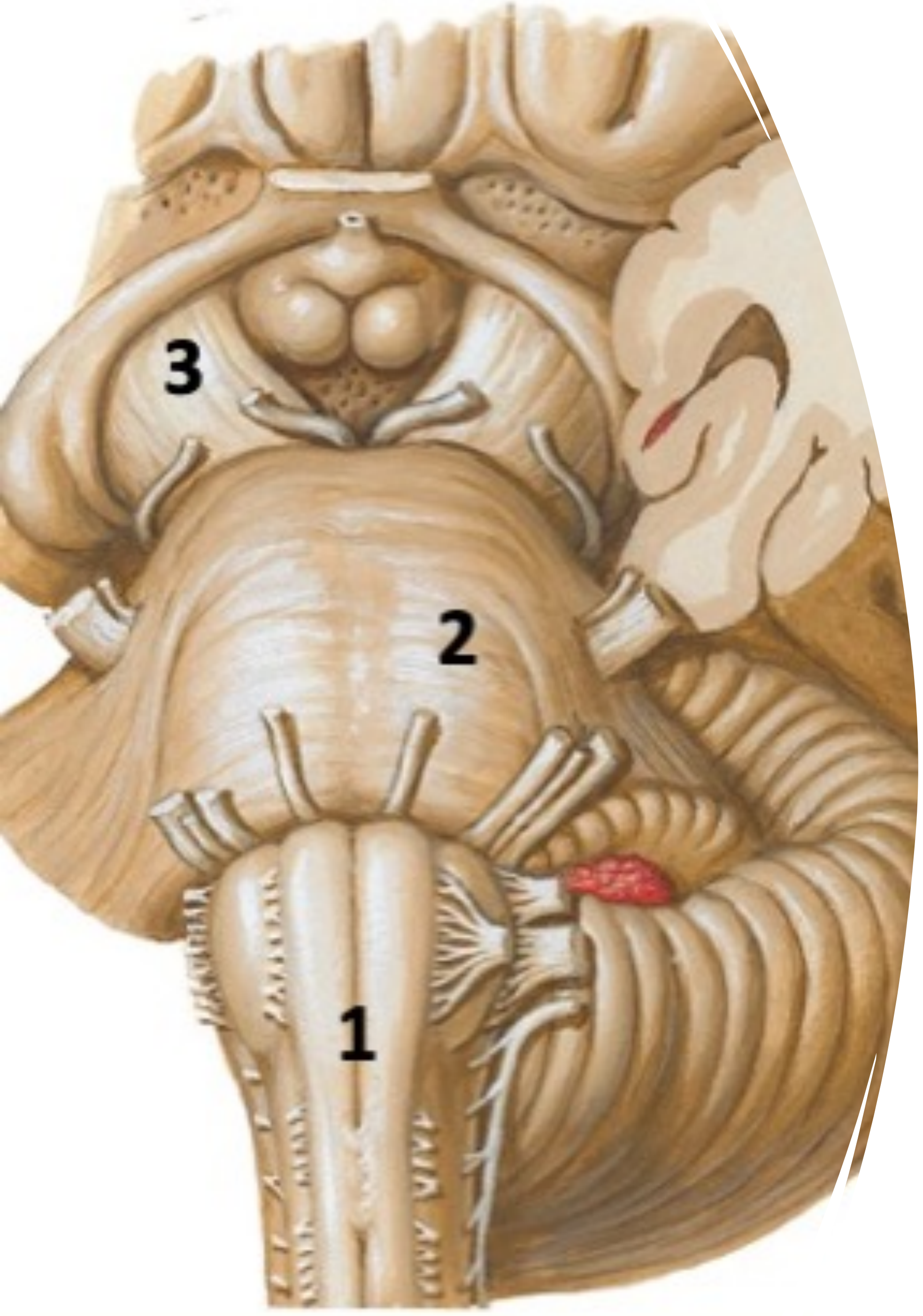


Cerebellum

- The cerebellum located in the posterior cranial fossa, behind the brain stem.
- It is composed of
  - Cerebellar hemispheres
  - Vermis
- Function: coordination of movements of the body







Brainstem

connects the narrow spinal cord with the expanded forebrain.

It consists of:

1. Medulla oblongata,
2. Pons
3. Midbrain

The brain stem contains:

- Nuclei of cranial nerves
- Descending tracts from brain
- Ascending tracts to brain
- Reticular formation

The 12 cranial nerves are:

1. Olfactory nerve.
2. Optic nerve
3. Oculomotor
4. Trochlear.
5. Trigeminal
6. Abducent.
7. Facial nerve.

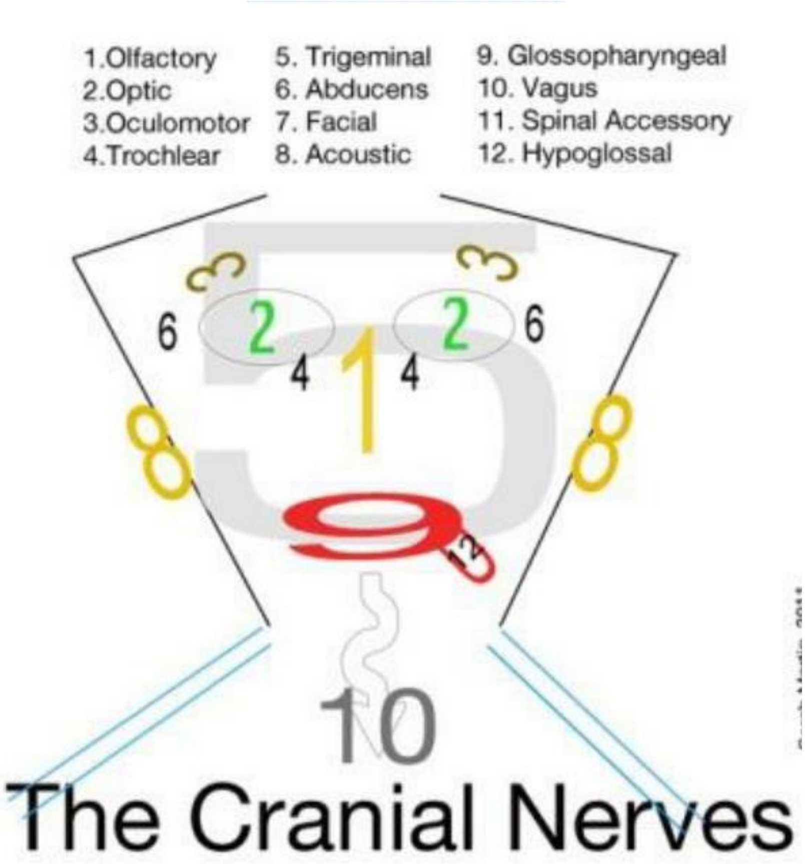
8. vestibulocochlear

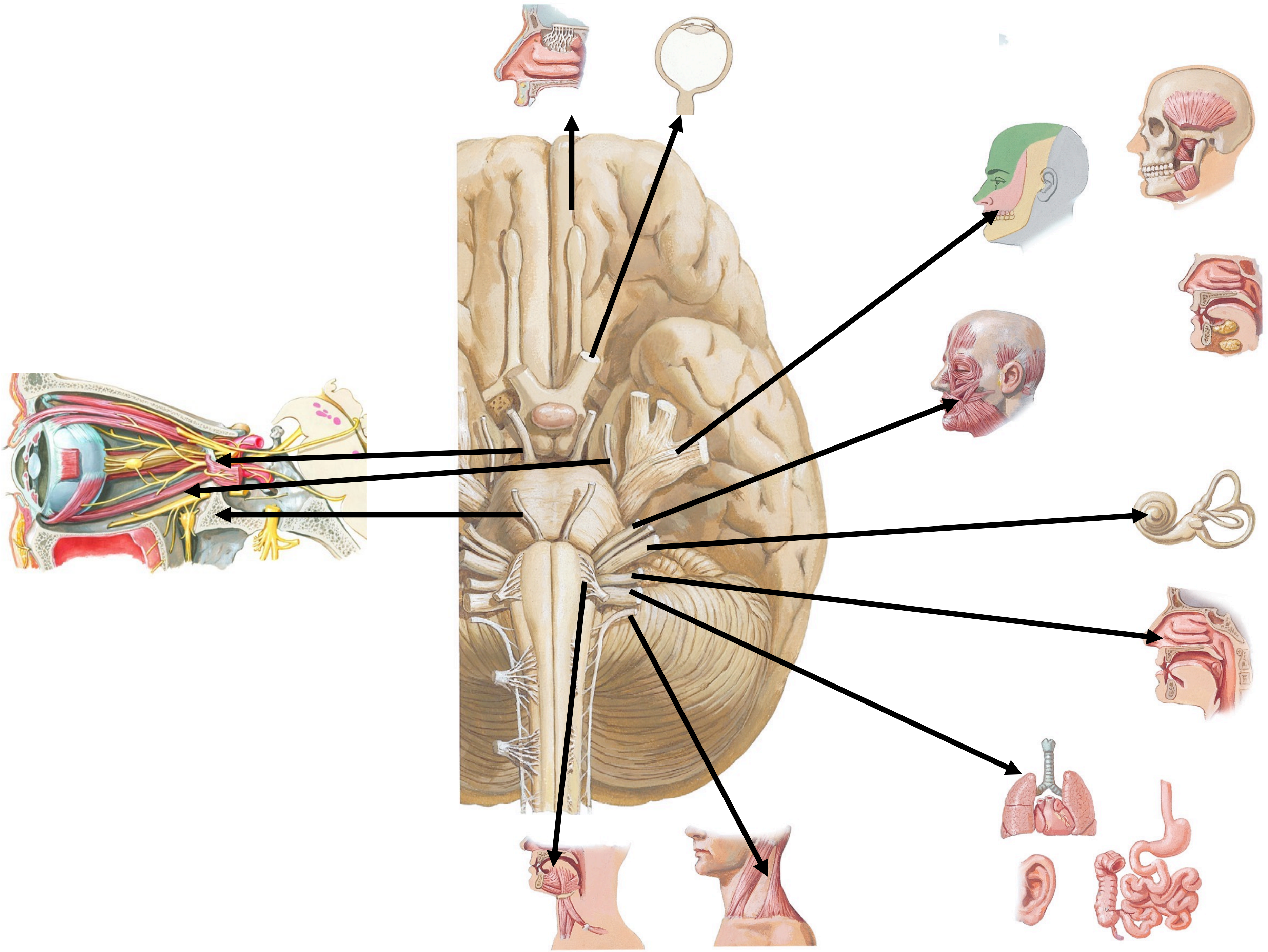
9. glossopharyngeal

10. vagus.

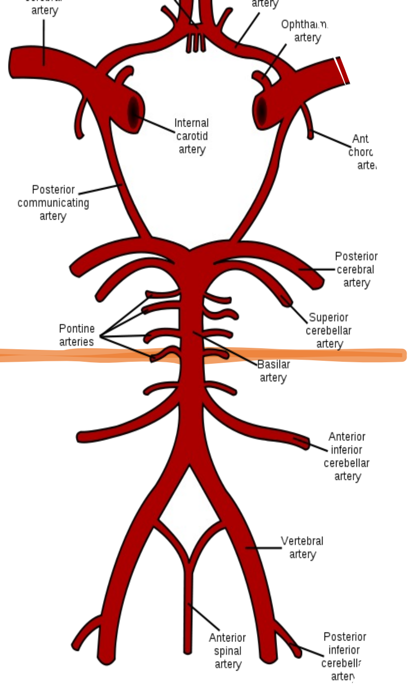
11. accessory

12. hypoglossal nerve.





Blood supply to the brain

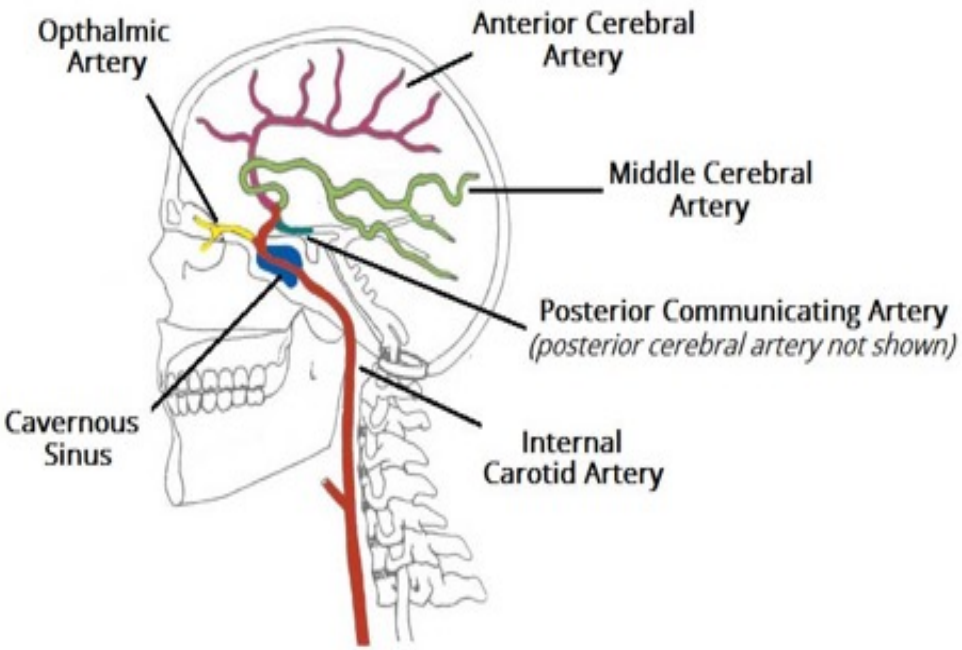


1. **<u>Internal carotid artery:</u>** divides to:

➢ Anterior Cerebral Artery(ACA)

➢ Middle Cerebral Artery(MCA)

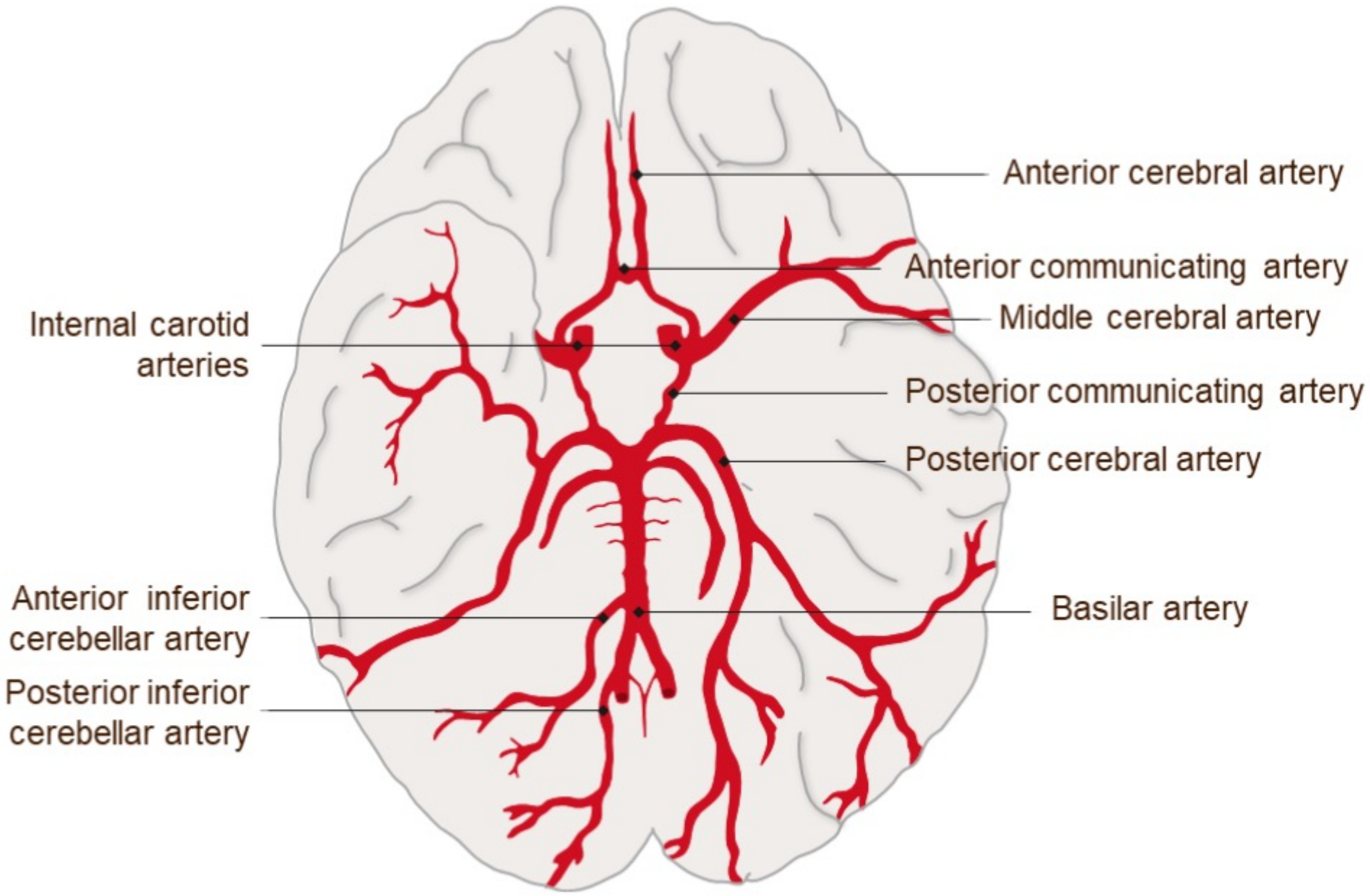
2. **<u>Vertebral artery:</u>** they join to form Basilar Artery. Then the Basilar artery divides to two Posterior Cerebral arteries(PCA).



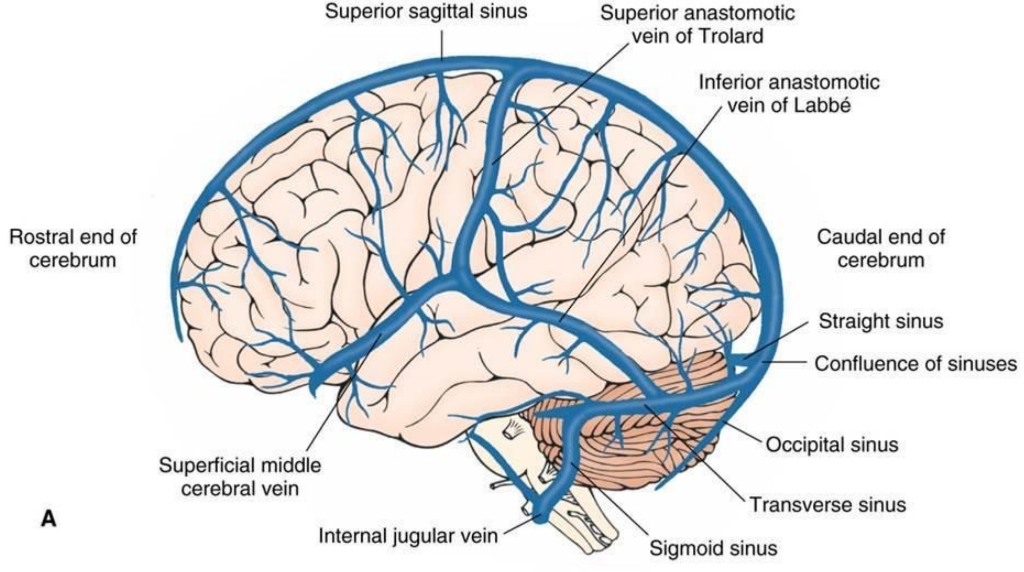
Circle of Willis

The internal carotid arteries and posterior cerebral arteries joined by Posterior communicating artery to form the Circle of Willis.

Arteries of the brain



Venous Drainage of the Brain



- Superficial cerebral veins Deep cerebral veins and Venous sinuses:

1. Superior sagittal sinus. 2. Inferior sagittal sinus 3. Straight sinus

4. Transverse sinus 5. Sigmoid sinus

They drain into internal jugular vein

Thank You